Paraphrase the given incomplete statement without changing the meaning. The same completion in the original input must also work for your paraphrase. Provide as many distinct paraphrases as you can come up with.

INPUT: The war during which Mario Stoppani was in the armed forces was
1) Paraphrased: The war Mario Stoppani was in the army in was
2) Paraphrased: Mario Stoppani was in the war called
3) Paraphrased: Mario Stoppani served in the army during

INPUT: The birth date of Luca Pianca is
1) Paraphrased: The date of birth of Luca Pianca is
2) Paraphrased: Luca Pianca was born in
3) Paraphrased: The day of Luca Pianca's birth is
4) Paraphrased: Luca Pianca's birth date is
5) Paraphrased: The birthday of Luca Pianca is

INPUT: The name of the architect of Ravenna Cathedral is
1) Paraphrased: Ravenna Cathedral was built by
2) Paraphrased: The person who built Ravenna Cathedral was
3) Paraphrased: The architect of Ravenna Cathedral is
4) Paraphrased: The architect behind the construction of Ravenna Cathedral is
5) Paraphrased: Ravenna Cathedral was built by the architect

INPUT: {prompt}
1) Paraphrased:

Figure 2: The few-shot prompt we use for paraphrase generation. We selected in-context examples from COUNTERFACT.

Editor	Score	Efficacy	Generalization	Locality
— (GPT-J)	37.4	34.4 (1.7)	34.5 (1.5)	45.3 (0.9)
ROME	35.0	39.8 (2.2)	25.5 (1.4)	**46.9** (0.8)
MEMIT	67.3	99.2 (0.3)	80.2 (1.2)	45.3 (0.8)
FT	55.8	68.1 (1.8)	60.4 (1.7)	44.5 (0.8)
FT + M + P + R	**68.5**	**99.6** (0.2)	**84.6** (1.1)	45.8 (0.9)

Table 8: Mass-editing results on WikiRecent (1,266 edits).

neighborhood prompts from COUNTERFACT from the corresponding train/test splits. The training data for our fine-tuning method consists of 1,266 edit requests, 18,990 augmented paraphrases (i.e., randomly generated), and 12,660 random neighborhood prompts from COUNTERFACT (not overlapping with those in the test set).

Table 8 shows the results. We again find that while vanilla fine-tuning is not effective, our conditional fine-tuning with random data augmentation is competitive with MEMIT.

B Generative Metrics for Single-Editing

For completeness, we give generative performance for single-editing in Table 9 as an accompaniment to Table 4. Our method is again competitive with the best performing single-editing baseline (ROME) in the edit score, but has lower generative scores.

Editor	Score	Fluency	Consistency
— (original GPT-2 XL)	30.5	626.6 (0.3)	31.9 (0.2)
FT	65.1	607.1 (1.1)	40.5 (0.3)
FT-L	66.9	621.4 (1.0)	37.4 (0.3)
KN	35.6	570.4 (2.3)	30.3 (0.3)
KE	52.2	586.6 (2.1)	31.2 (0.3)
MEND	57.9	624.2 (0.4)	34.8 (0.3)
ROME	89.2	621.9 (0.5)	41.9 (0.3)
FT + Mask + Para + Sim	83.1	557.0 (1.8)	15.2 (0.3)

Table 9: Fluency and consistency with single-editing on COUNTERFACT.

C Data Augmentation for SWEA⊕OS

We train SWEA⊕OS on a set of random facts $\mathcal{R}_\mathcal{E}$ in addition to random paraphrases (which is already present). We use 10,661 random facts containing unique subjects from $\mathcal{E}$ within the training split. This is done because SWEA⊕OS is only concerned with a final cache containing subject tokens and their embeddings. We first obtain the subject embeddings of the requested edits in $\mathcal{E}$ as described in Li et al. (2024a). We then repeat the process for the augmented random facts in $\mathcal{R}_\mathcal{E}$. Between the two sets of subjects, there are 392 subjects that overlap. In the case of an overlap, we only retain the embeddings from the $\mathcal{E}$ and remove those from $\mathcal{R}_\mathcal{E}$.